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Multi-railway Track and Switch Region Recognition Using Mobile Laser Scanning Data

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***Abstract*—Railway infrastructure is a vital component of modern transportation systems. However, today's railway equipment inspections primarily rely on labor-intensive fieldwork and error-prone human visual analysis. In this study, we present an automated system for the recognition of railway assets using mobile laser scanning data. Our method simultaneously detects rail tracks and localizes switch regions, identifying switch region orientation (right/left), status (open/closed), and overlap type (merge/split). This integrated railway tracing and switch recognition method is built on binary multiscale template matching within a Kalman filtering framework. To trace the railway trajectories, the Kalman filter predicts a new state vector that consists of the position and orientation of the rail track window. This prediction which is based on the previous state vector is then updated using the observed track points already classified in a local track window using the Gaussian mixture model clustering method. To detect multi-tracks and determine appropriate observations for the targeted rail track, we employ minimum description length technique. Lastly, to recognize the switch type, template matching is applied by considering the similarity between a multi-track region and the templates as well as track design constraints observed in the sequential multi-track regions. We test our algorithm on a densely-annotated private benchmark of 3D point cloud data captured from a large-scale railway network. Experiments on this dataset show that the proposed method can robustly produce accurate rail track vectors (99.46% recall) and recognize switch types (97% success rate).**

I. Introduction

Railway infrastructure is a crucial asset of modern transportation systems. Each year, railways carry around 20 billion passengers and 10 billion tons of freight worldwide, and the global demand for railway development was projected to grow significantly by 2028 [1]. Modern railway systems like communication-based train control (CBTC) rely on precise geospatial data on their infrastructure to operate smoothly [2]. Accordingly, rail track (equivalent to a pair of parallel rails creating a rail path) is a key object that has a direct effect on railway operation [3]. Thus, regular inspection is required to ensure appropriate maintenance of the rail tracks. However, railway inspections still depend primarily on labor-intensive fieldwork and error-prone human visual analysis [4].

Recently, mobile laser scanning (MLS) technology has been rapidly emerging as an alternative surveying tool. This surveying system mounts a light detection and ranging (LiDAR) scanner on a moving vehicle, which is tightly coupled with navigation sensors such as inertial navigation system (INS) and global positioning system (GPS) [5]. This surveying system captures 3D object surfaces with billions of laser points acquired along a typical urban railway [6].

Although previous works have used MLS data for automatic rail detection [7,8,9,10,11], they generally stop at segmentation and do not delineate individual rail instances or pair left and right rails into complete path instances. Furthermore, in spite of their acceptable performance within single-track scenes, rail tracks in multi-track scenes are not sufficiently dealt with due to the inherent complexity. However, in railway systems, switch regions with multiple tracks where rail tracks split (turnouts) or merge are crucial for safe railway operation.

Another research gap is that, despite the numerous advantages of LiDAR surveying, the huge data volume makes the processing extremely time-consuming [12]. To address this, different downsampling methods can be employed before analyzing data. Alternatively, data extraction from the point cloud can be automated, which can significantly reduce the overall cost of LiDAR scanning [13]. Therefore, our paper proposes an automated method to recognize railway assets using 3D MLS data. The main contributions are summarized as follows.

- Using a 3D point cloud captured via multiple scans within a railway network, our method accurately detects left and right rail points of each rail path instance through constructing multi-scale spatiotemporal windows drawn over different classes of 3D rail points.
- Regardless of the railway network length, we trace the rail paths within the object space through Kalman-based vector representation.
- The proposed algorithm can reliably localize switch regions and determine their types using minimum description length (MDL) algorithm and binary multi-scale template matching while considering spatiotemporal relations between multi-track regions.

A. Related Works

Recently, there has been an increasing interest in automatically extracting or modeling rail tracks from remotely sensed data [14]. Initial efforts mainly focused on detecting rails and tracing rail paths on images for driver assistance systems. Track bed (rail region) segmentation has been performed using the Hough transform [15], or single-rail hypotheses through linear edge extraction [16]. Rail detection by iteratively selecting local maxima of the gradient image [17] has also been performed. A subsequent approach employs the histogram of

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oriented gradients (HOG) for rail detection, then uses a mirror-based technique to generate a virtual rail for switch recognition [18].

In terms of image-based methods that use deep learning (DL), two recent studies have used DeepLabv3+ [19] and RailNet equipped with depth-wise convolution [20] to semantically segment the rails. Another line of work [21] pairs the semantically segmented rails using key points and topological features of the scene. Another study uses D-LinkNet [22] to segment the track bed while employing YOLOv3 [23] to detect the foreign objects. BiSeNet [24] is also proposed to segment rails and track beds for detecting railway anomalies. Image-based rail understanding is vulnerable to background variation, shadow and occlusion noise, and fails in low-light conditions (night or tunnels). Moreover, recent studies mostly yield only 2D data, whereas precise 3D information is essential for geometric inspection. [25].

Regarding rail scene understanding using MLS data, a study models railway vectors by detecting track points based on geometric features and the Fourier curve [6]. Reference [7] detects track beds by analyzing consecutive cross sections while extracting track points using geometric shape and intensity features. Another study [8] extracts rails by considering height variation on the track bed, continuity, and smooth curvature gradient. Other studies [9,25] model a track as a dynamic system of local pairs of parallel line segments in the Kalman filter framework. Recently, several research works have proposed methods to detect rail tracks in multi-track scenes. Reference [4] detects tracks, closure rails, and frogs using feature-based methods, support vector machine (SVM), and an iterative closest point (ICP) algorithm. Rail extraction using template matching and spatial clustering is another line of research [5]. However, the two aforementioned methods [4,5] require information using a single scan to detect and trace rail track points.

In terms of MLS-based methods using DL, a recent study [13] filters the point cloud, processes the points into point-voxel pairs, extracts voxel-wise features through a neural network, and finally predicts each voxel class. FarNet [26] is also proposed to detect tracks. In FarNet, a point attention mechanism combined with an information aggregation module makes the network focus on more valuable pixels. Since the two previous methods [13,26] produce sparse segmentations, a density-independent approach that takes advantage of point cloud spatial transformations was recently introduced [27]. However, this method does not associate left and right rail points to form complete tracks, a capability essential for the automatic survey of railways.

II. Rail Track Extraction

Rail track is an important object in railway systems when considering efficient railway operation and safety. In this context, a switch is a mechanical setup on the rails that guides the train to change or maintain its path. Rail frog is a designed rail piece at the intersection of two rails, creating a gap through which a wheel can pass smoothly. Toe end of a frog is located at the end of a frog close to the switch point. Guard rails are also occasionally constructed parallel to the rail tracks on the track bed to avoid derailment.

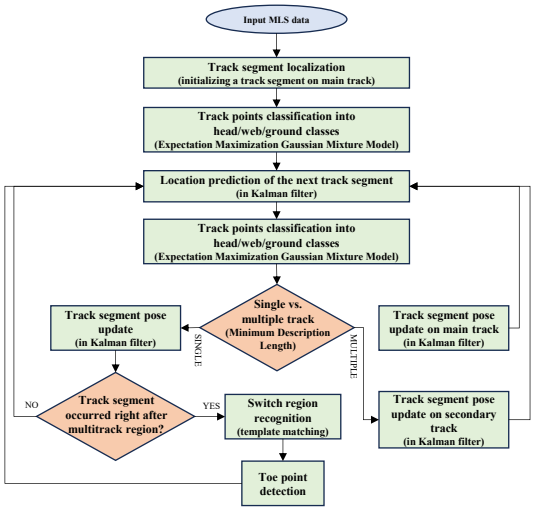


Fig. 1. Flowchart of the proposed joint recognition of tracks and switches.

Fig. 1 illustrates the flowchart of our proposed method for jointly detecting and tracing rail tracks and switch regions using MLS data. A track segment comprising a pair of track windows is designed to represent a pair of rails on a local scale (Section II-A). The rail track extraction process within the whole railway starts by manually setting the position and orientation of the starting pair of track windows. Points belonging to a track window are classified by considering the height distribution. Then, a Kalman filter is applied to trace the trajectories of rails (Section II-B). To detect multi-tracks, a model selection process is conducted by applying the MDL technique (Section II-C). Also, for switch regions, the region type is recognized using binary multi-scale template matching (Section III).

A. Design of Multi-scale Track Windows

Rail tracks and switches have specific configurations that can be observed at different scales. In this study, we design a multi-scale and spatiotemporal window to facilitate joint recognition of rail tracks and switch regions. The proposed multi-scale and spatiotemporal approach consists of different scales. At the smallest scale, we define a track window W which is used to analyze MLS points along a single rail by placing it in the center of the rail having the same orientation as the rail. As shown in Fig. 2, W has three sub-windows: $W = \{W_m, W_s^{\text{left}}, W_s^{\text{right}}\}$. Each sub-window is designed to analyze different parts of W where the main sub-window W_m focuses on the rail itself, whereas the two side sub-windows W_s^{left} and W_s^{right} collect points from the left and right sides of the rail, respectively. The rail direction determines the left or right location of the side sub-windows. The length of these sub-windows is heuristically set as 1 m, and the width is set as 0.20 m. 3D track points of the current rail path are likely to be observed in the main sub-window, whereas track points

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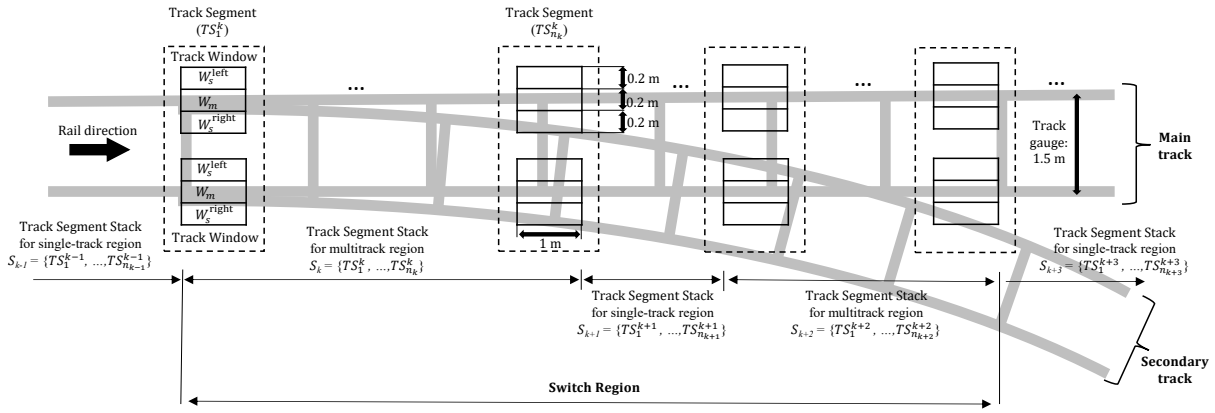


Fig. 2. Illustration of multiscale spatiotemporal windows, i.e., track window, track segment, and track segment stack within a sample switch region comprising of a main track and a secondary track. The switch region here starts with a multi-track region followed by a single-track region, and ends with another multi-track region.

from other paths (observed in multi-track regions) are likely to be observed in the side sub-windows.

On an intermediate scale, we introduce a track segment TS as a unit for railway trajectory tracing, which consists of two parallel track windows. A track segment is designed to deal with the parallelity of rails within a rail track. The length between the centers of two parallel track windows (track gauge) is set to 1.5 m considering the regularity of the railway system. A track segment is divided into either a single-track segment or a multi-track segment based on the presence of multiple rail tracks. If multiple rails are observed in one or both track windows belonging to a track segment, the track segment is considered to be a multi-track segment.

On a larger scale, we define the track segment stack $S_k = \{TS_1^k, ..., TS_{n_k}^k\}$, with n_k being the number of track segments in S_k. The track segment stack S_k represents a spatiotemporal region where single-track or multi-track segments are continuously observed. Thus, the entire railway scene R is made up of N track segment stacks where $R = \cup_{k=1}^{N} S_k$ and $S_i \cap S_j = \emptyset$ if $i \neq j$. Note that a track segment stack consisting of only multi-track segments is used for switch recognition, according to Section III. Fig. 2 further elaborates the concept of our multi-scale spatiotemporal method.

B. Rail Track Vector Extraction using Kalman Filter

In this study, we use the Kalman filter [28] to model the gradually changing pose of rail tracks as a dynamic process considering a distance along a rail track instead of a time epoch represented in previous research [9,25]. The process consists of predictions and updates. In the prediction stage, the position vector $\vec{X}_q = [x_q \ y_q \ z_q]$ for the current track window is predicted based on the previous track window position $\vec{X}_{q-1}$ [9], as mathematically illustrated in (1).

$$[x_q \ y_q \ z_q] = [x_{q-1} \ y_{q-1} \ z_{q-1}] + \Delta L[\sin\theta_{q-1} \cos\theta_{q-1} \ 0] \quad (1)$$

Here, $\vec{X}_q = [x_q \ y_q \ z_q]$ is a state vector representing the center position (x, y, z) of the track window. θ_{q-1} and ΔL are the orientation and length (1 m in this study) of the previous track vector $\vec{X}_{q-1}$, respectively. The z component remains constant assuming that the sequential rail track vectors remain at the same height at a local scale.

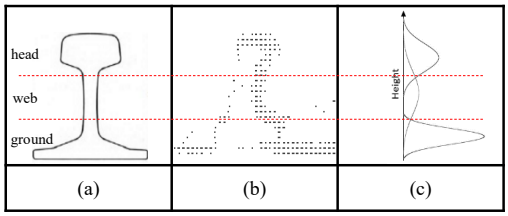


Fig. 3. Configuration of the MLS points inside a track window: (a) track model, (b) MLS data, and (c) height distribution.

Once a track window is predicted from the prediction stage, track head points, which are used as the observation in the update stage of the Kalman filter, are detected based on the expectation maximization-Gaussian mixture model (EM-GMM) clustering method. Although MLS points are acquired from various objects of a railway scene, points in a track window mainly contain specific objects including ground and rail. Thus, points belonging to a track window can be clustered into track head, track web, and ground (including track bed), each having unique geometric characteristics with a certain height distribution, as shown in Fig. 3.

We assume that the points belonging to a track window have three class labels $C_i \in \{\text{head}, \text{web}, \text{ground}\}$. We also assume that the points belonging to each class follow a Gaussian distribution $\mathcal{N}(\mu, \sigma)$ in terms of height, with expectation μ and variance σ. Then, the GMM with three components can be formulated as (2).

$$p = \sum_{c=1}^{3} \alpha_c \mathcal{N}(\mu_c, \sigma_c) \quad (2)$$

Here, p is the height distribution, and α_c represents the mixture

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weights where $\sum_{c=1}^{3} \alpha_c = 1$. The GMM parameters can be optimized in the EM framework. Once the GMM parameters are determined by the EM algorithm, points are clustered into three classes by comparing probabilities of the three Gaussian distribution components. Points belonging to the component with the highest mean are considered track head points which are subsequently used as the observation for rail track extraction in the update stage of the Kalman filter.

During the update stage, we use the center of track head points extracted within the EM-GMM framework to update the predicted state vector according to (3).

$$\vec{X}_q = \vec{X}_q + K(\vec{O}_q - \vec{X}_q) \tag{3}$$

Here, *K* is the Kalman gain and $\vec{O}_q$ is the observation. $\vec{O}_q$ is the average of track head points coordinates in the track window W_q. In (3), *K* controls the significance between the predicted state vector and the observed state vector. As *K* increases, the next estimated state vector is more affected by the observed state vector. The value of *K* is optimized by considering the prediction noise covariance matrix and the measurement noise covariance matrix so that the estimated state vector minimizes the covariance matrix of the estimated error [29].

C. Model Selection Method to Detect Multi-track Segments

The recognition of multi-track regions from track points detected using the method of Section II-B is not easily achieved due to the unknown number of existing rail tracks. In single-track regions, track head points extracted in the track window can be considered as observation, whereas in a multi-track region where two rail tracks merge or split, track head points are reflected from more than one rail. Thus, the track head points belonging to the main track should be separated from the others if a multi-track segment is observed. Then, the update stage of the Kalman filter should be separately performed for each detected track.

To recognize a multi-track segment, we determine the number of rails and their locations, which are subsequently used for switch recognition. Also, track head points belonging to a targeted rail are determined, and then, used as observations for the Kalman filter. In this study, we adopt the MDL to detect multi-track segments [30,31]. Thus, according to (4), the description length function $L(D)$ consists of a model complexity term and a model closeness term.

$$L(D) = \lambda L(D|H) \ + \ (1-\lambda)L(H) \tag{4}$$

Here, $L(D|H)$ indicates a goodness-of-fit of observation *D* given a model *H*, whereas $L(H)$ represents how complex the model *H* is. Consequently, $L(H)$ is related to the number of model parameters, whereas $L(D|H)$ corresponds to the weighted sum of the squared residuals between a model *H* and a set of observations *D*, i.e., $[D-H]^T[D-H]$ in matrix form. λ is a weight to balance the model closeness and the model complexity. In the current step, we aim to know how many rails occur in a track window. Thus, hypotheses are generated with different numbers of straight lines. Here, the number of lines is equal to the number of parameters. Straight lines to be used as a model are extracted from the track head points using the random sample consensus (RANSAC). In this study, we assume that maximum three rails can be observed in a track window. So, three hypotheses $H = \{h_1, h_2, h_3\}$ are generated for a track window. The optimal model H^* is selected according to (5) by minimizing the $L(D)$ value among the competing hypotheses.

$$H^* = \arg\min_{\forall H} \{\lambda L(D|H) \ + \ (1-\lambda)L(H)\} \tag{5}$$

If the optimal model has one straight line, the track window lies in a single rail region. Otherwise, the track window is recognized as a part of a multi-track region. It should also be noted that the updated track window takes the orientation of the straight line found through the MDL.

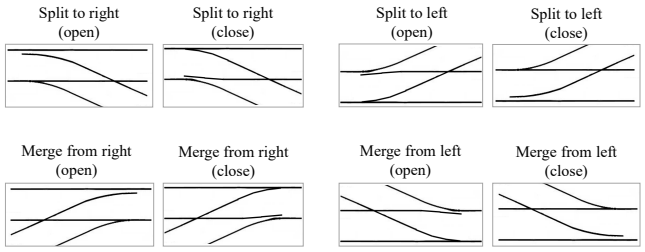


Fig. 4. Different types of railway switch regions. The horizontal rail tracks are the main tracks whereas the curved ones represent the secondary tracks that either merge in or split out.

III. Switch Region Recognition

Switch regions can be classified into eight categories depending on the overlap type (merge/split), direction of the secondary rail track (left/right), and the location of the toe (open/close), as shown in Fig. 4. In this section, we recognize the switch region type by comparing the configurations of its multi-track regions with some designed templates.

A. Template Design

A switch region incorporates one multi-track region at the start and one at the end, with a single-track region lying in between the two, as illustrated in Fig. 2. A multi-track region is essentially a track segment stack generated by a series of multi-track segments. Likewise, a single-track region is essentially a track segment stack generated by a series of single-track segments. Here, a multi-track region is represented by two multi-track segments at the start, and two at the end of the region.

In order to understand the switch type, we have to recognize the two multi-track regions included at the start and end of the switch region. A mechanical switch can take place at either the start or end of a switch region. Each switch has one multi-track region at its location and a second multi-track region positioned either directly in front of or directly behind it. The multi-track region that covers the mechanical switch can have eight different types, while the other multi-track region

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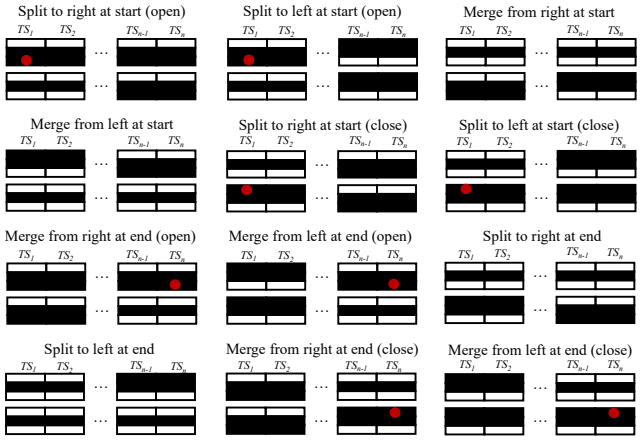


Fig. 5. Templates for multi-track regions inside a switch region (black: occupied by a rail and white: not occupied by a rail). Also, toe point for the multi-track region corresponding to the switch is visualized with a red dot.

of the switch region can have only four configurations. For instance, according to Fig. 5, in the case of "split to right" switches, the mechanical switch and its corresponding multi-track region take two different types (open or close), whereas the multi-track region occurring at the end of the switch region has a single configuration. Therefore, the multi-track regions can have 8 + 4 = 12 different templates $T_{m=1,\ldots,12}$. These templates are differentiated based on the overlapping type (merge/split), direction of the overlap (left/right), location of the toe (open/close), and the location of the multi-track region in the switch region (start/end). As shown in Fig. 5, rails are ideally observed in certain track sub-windows (black fill represents the presence of rail). Also, the approximate positions of the toes are ideally determined (red dot). However, the length of the track segment stack is unknown and changes depending on the track design such as curvature of the track. That is why the multi-track region understanding process focuses on four specific track segments in the track segment stack (the first two and the last two track segments). Then, each template is converted into a binary string of length 24 which is a sequence of zeros (not occupied by a rail) and ones (occupied by a rail). As an example, "split to right at start (open)" in Fig. 5 is expressed as {0,1,1,0,1,0,0,1,1,0,1,0,0,1,1,0,1,1,0,1,1,0,1,1}.

B. Switch Type Recognition

Let S_k and T_m denote a track segment stack and a template, respectively. Similar to the templates, a track segment stack is represented by a 24-bit long string based on four attributes, i.e., the overlapping type, the direction of overlap, the location of the toe, and the location of S_k in the switch region. Then, multi-track region type recognition is carried out using a score function that consists of a unary term and a contextual term according to (6).

$$Score(k,m) \;=\; w.U(S_k,T_m) + (1-w).C(S_{k'},S_k) \quad (6)$$

Here, ω is a weight for balancing the unary term U and the contextual term C, which is heuristically set to 0.9 in this study.

1) Unary Term: This term measures the similarity between binary strings for a multi-track region S_k and the template T_m. Taking binary values for i and j, the indicator function $\delta_e^{k,m}(i,j)$ is calculated as (7).

$$\delta_e^{k,m}(i,j) = \begin{cases} 1 & S_k(e)=i \text{ and } T_m(e)=j \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Here, $1 \le e \le 24$, where $S_k(e)$ and $T_m(e)$ are the eth element of the binary string for S_k and T_m, respectively. Then, the occurrence number $n_{ij}^{k,m}$ in a binary string of length 24 is calculated as (8).

$$n_{ij}^{k,m} \;=\; \sum_{e=1}^{24} \delta_e^{k,m}(i,j) \quad (8)$$

From (8), four different matches $(n_{00}^{k,m}, n_{01}^{k,m}, n_{10}^{k,m}, n_{11}^{k,m})$ are generated. $n_{00}^{k,m}$ and $n_{11}^{k,m}$ denote negative and positive matches, respectively, whereas $n_{01}^{k,m}$ and $n_{10}^{k,m}$ denote mismatches. Based on four different matches, the similarity measure is formulated as (9).

$$U(S_k,T_m) = \frac{n_{00}^{k,m} + n_{11}^{k,m}}{n_{00}^{k,m} + n_{01}^{k,m} + n_{10}^{k,m} + n_{11}^{k,m}} \quad (9)$$

2) Contextual Term: This term is designed to consider the relationship between track segment stacks at two different time epochs (a current track segment stack S_k and a previous track segment stack $S_{k'}$) as (10). Here, the type of $S_{k'}$ is already known and matches its corresponding template.

$$\begin{aligned} C(S_k,S_{k'}) \;=\; & \delta_{\text{Start/End}}(k,k') + \\ & \delta_{\text{Start/End}}(k').[\delta_{\text{Merge/Split}}(k,k') + \delta_{\text{Left/Right}}(k,k')] \end{aligned} \quad (10)$$

As shown in (10), the contextual term consists of a combination of four indicator functions that represents the compatibility between labels of the previous known multi-track region and a current multi-track region. Generally, multi-track regions representing the start or end of a switch region are alternately observed in a railway scene. Thus, $\delta_{\text{Start/End}}(k,k')$ is designed to favor different attributes for two sequential multi-track regions as (11).

$$\delta_{\text{Start/End}}(k,k') = \begin{cases} 1 & \text{Start/End attribute is different} \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

Two sequential multi-track regions are likely to have the same labels for the merge/split attribute and left/right attribute if the two belong to the same switch region. Thus, $\delta_{\text{Merge/Split}}(k,k')$ and $\delta_{\text{Right/Left}}(k,k')$ favor the same attributes whereas $\delta_{\text{Start/End}}(k')$ is designed to decide whether or not a current multi-track region makes a pair with a previous one, as shown in (12)-(14).

$$\delta_{\text{Merge/Split}}(k,k') = \begin{cases} 1 & \text{Merge/Split attribute is the same} \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

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$$\delta_{\text{Right/Left}}(k,k') = \begin{cases} 1 & \text{Right/Left attribute is the same} \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

$$\delta_{\text{Start/End}}(k') = \begin{cases} 1 & S_{k'} \text{ is Start} \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

Among 12 templates, the optimal type is determined by maximizing the designed score function $Score(k,m)$ with respect to m. Switch region recognition is then achieved as the two multi-track regions of the switch region are successfully matched with their corresponding templates. Once an input switch region is matched, an approximate position of the toe can also be determined. Accurate position of the toe is determined by collecting track head points belonging to a window marked as the position of the toe, and then, finding the end point of the track head points in the rail direction.

IV. Experimental Results

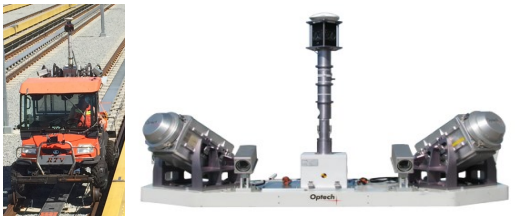
A. Datasets

The proposed algorithm was tested on MLS data provided by Thales Canada. The MLS data were acquired in 2016 using Teledyne Optech's Lynx SG1 Mobile Mapper system mounted on an inspection train in Vancouver, Canada. The range precision is 5 mm, absolute accuracy is ±5 cm, laser measurement rate is 150-1200 kHz, and the maximum range is 250 m. The dataset contains two main tracks (paths 1-2), a parking zone which contains five tracks (path 3-7), two sub-tracks (path 8-9) which run between other tracks, and a hundred selected multi-track regions across the railway network. We use these extra multi-track regions to evaluate multi-track recognition performance and conduct ablation studies. The length of the main tracks is approximately 1.58 km, and a total of 11 switches (22 multi-track regions) is observed within the paths 1-9. Fig. 6(a) shows the Lynx SG1 Mobile Mapper system, while Fig. 6(b) shows the parking zone point cloud of the Vancouver dataset with the ground truth track head points of paths 1, 2, 3, 6, and 7 annotated in red.

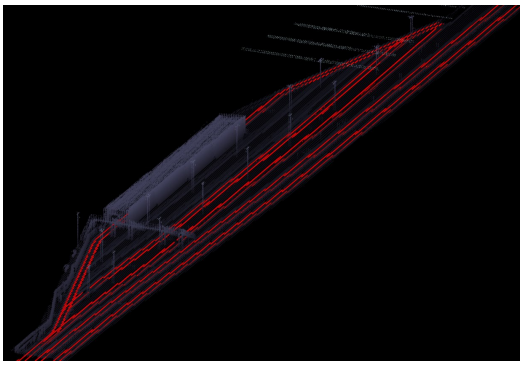
B. Rail Tracing Results

After initializing the railway trajectory by setting the pose of the initial track segment, rail track vectors (equivalent to the track segments) are automatically extracted using the proposed railway tracking method. Rail track vectors were extracted for 9 paths as shown in Fig. 7. Vectors in path 3, path 6 and path 7 were broken due to occlusions where a very small number of track points were observed (circles in Fig. 7 show the broken regions). Fig. 8 shows the enlarged rail track vectors where the rectangles represent track windows, with blue and red colors denoting single-track and multi-track regions, respectively. Visual inspection indicates that multi-tracks as well as single tracks are well extracted along with the rail track points.

In order to quantitatively evaluate the accuracy of the extracted rail track vectors, track points were manually classified using a commercial software (TerraScan). We first computed point-to-model distance in horizontal and vertical directions. Point-to-model distance measures the average distance between the extracted track vector (straight line found by MDL) and the ground truth track points. As shown in Table I, the average distances in horizontal and vertical directions are 2.5 cm and -0.3 cm, respectively, with standard deviations of ±2.3 cm and ±0.8 cm, respectively. The result indicates that the extracted rail track vectors fit well with the corresponding track head points, considering that the width of the track head is approximately 12 cm. The point-to-model distances for path 8 and path 9 are relatively larger than the others, because those paths are relatively short and contain multi-track regions that cover large portion of the paths. In switch regions, track points are not continuously observed on one side of parallel rails. Thus, extracted track vectors are smoothly transited while point-to-model distance increases due to the discontinuous track points.



(a)



(b)

Fig. 6. (a) To gather MLS data of Vancouver dataset, an inspection train shown on the left carries the Lynx SG1 mobile mapper system which is shown on the right. (b) 3D point cloud of the parking zone in Vancouver dataset. To give a clear visualization, the red dots show the ground truth track head points of only paths 1, 2, 3, 6, and 7.

TABLE I
Quantitative Assessments for Rail Track Extraction

Path	Length (m)	Point to model distance (cm)				Recall (%)	Description
		Horizontal		Vertical			
		Avg.	Std.	Avg.	Std.		
1	1581	1.9	±1.3	-0.2	±1.0	99.95	Main track
2	1578	1.9	±2.9	-0.3	±1.0	99.92	Main track
3	338	2.8	±2.4	-0.3	±0.5	99.23	Parking lot
4	181	2.8	±2.5	-0.4	±1.0	99.34	Parking lot
5	213	2.6	±2.3	-0.4	±1.0	99.52	Parking lot
6	240	2.5	±2.0	-0.4	±1.1	99.90	Parking lot
7	264	2.0	±1.6	-0.3	±0.8	99.97	Parking lot
8	47	3.0	±2.5	-0.2	±0.2	99.19	Subtrack 1→2
9	47	3.1	±2.8	-0.2	±0.5	98.15	Subtrack 1→3
Total	**4489**	**2.5**	**±2.3**	**-0.3**	**±0.8**	**99.46**	

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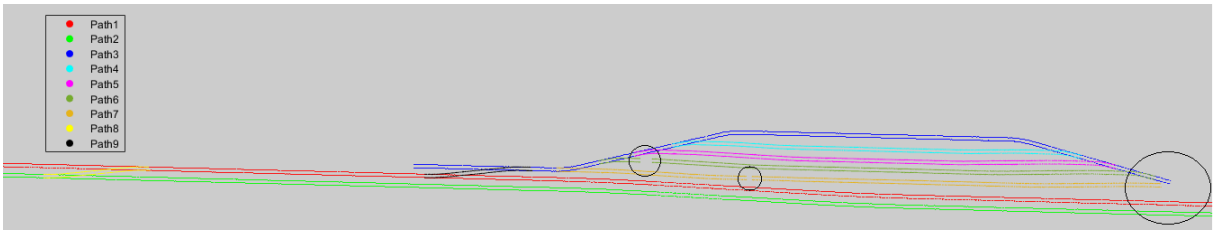


Fig. 7. Top view of the extracted rail track vectors for paths 1-9. Paths 1 and 2 which are the main tracks are much longer than the other paths. So here, only a small portion of these two paths that are close to the parking zone is visualized. Vectors in path 3, path 6 and path 7 are broken due to occlusions where a few number of track points were observed. The black circles show the broken regions.

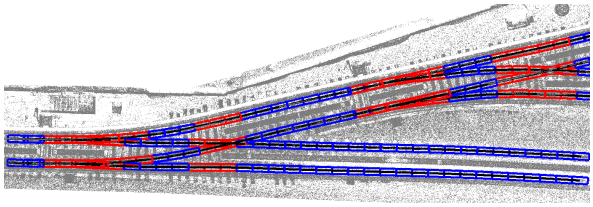


Fig. 8. Extracted rail track vectors for a sample switch region. Blue and red colors denote single-track and multi-track regions, respectively.

TABLE II
QUANTITATIVE ASSESSMENTS FOR MULTI-TRACK TYPE RECOGNITION

	Multi-track detection (%)	Multi-track type recognition			
		Correct	Multiple solution	Incorrect	Success rate (%)
Unary and contextual term ($\omega = 0.9$)	100	97	3	0	97
Unary term only ($\omega = 1$)	100	69	28	3	69

In order to compute the recall values of Table I, buffer zones with a width of 12 cm (corresponding to the track head width) were created for the extracted track vectors. A track point is considered a true positive if it is observed within the buffer zone. Otherwise, it is considered a false negative. The overall recall for all paths was 99.46%, as shown in Table I. Fig. 9 shows the true positives (black) and false negatives (red) visualized by color. The results indicate that a very high recall was achieved in single-rail scenes, even in curved regions. Most false negatives were observed either near the switch regions or regions with relatively high curvature. Thus, similar to the results for the point-to-model distance, relatively low recall was achieved in path 8 and 9.

C. Switch Type Recognition Results

During the rail tracing process, both single-track and multi-track regions should be detected. Although the results so far cover only 22 multi-track regions (11 switches), we test a hundred extra multi-track regions to evaluate the performance of the proposed switch type recognition method. According to the first row in Table II, all multi-track regions were successfully detected (100% detection rate). Here, 97 multi-track regions were correctly recognized compared with the ground truth type (97% success rate). For 3 multi-track regions, the method proposed multiple solutions where two template types had the same score values. The main reason for these multiple solutions is the matching ambiguity caused by noise observation. Structures such as guard rails and frogs attached to the rail tracks cause different occupancies in a binary string compared to a designed template. Also, rail track points that are partly observed in a track window cause occupancy errors.

Additionally, using the same aforementioned 100 multi-track regions, we evaluated the effect of the contextual term in the score function (as an ablation study) by setting the weight parameter ω to 1. According to the second row in Table II, out of 100 multi-track regions, 69 were correctly recognized. Yet, multiple solutions were proposed for 28 multi-track regions, and 3 multi-track regions were incorrectly recognized. The results show that the contextual term has a positive effect on resolving the matching ambiguity by reinforcing the contextual force.

Our immediate future work is to expand the experiments by evaluating our approach on publicly available LiDAR datasets such as Rail3D [11] or RailPC [32] to enable fair comparisons with other rail extraction studies. However, all these public benchmarks lack separate left/right rail annotations and do not include multi-track or switch type labels, which constrains the assessment of certain aspects of our contributions.

V. CONCLUSIONS

A quantitative assessment for point-to-model distance showed that the average distances between extracted rail track vectors and track points in horizontal and vertical directions were 2.5cm and -0.3cm, respectively, with an standard deviation of ±2.3cm and ±0.8cm, respectively. Furthermore, a 99.46% recall score was achieved. These results indicate that the proposed rail tracking method can extract reliable and accurate rail track vectors in both single-track and multi-track regions. All multi-track regions were well detected by the MDL-based model selection method. A 97% success rate for multi-track type recognition was also achieved. Moreover, the results showed that the contextual term plays an important role in robustly recognizing multi-track region type. In this study, the proposed method was applied to a dataset with a very dense point cloud. In future works, we will conduct a sensitivity analysis on other datasets with different properties, such as low-density point clouds. Additionally, we will extend our method by adding a probability value to templates in order to reduce the ambiguity of switch type recognition.

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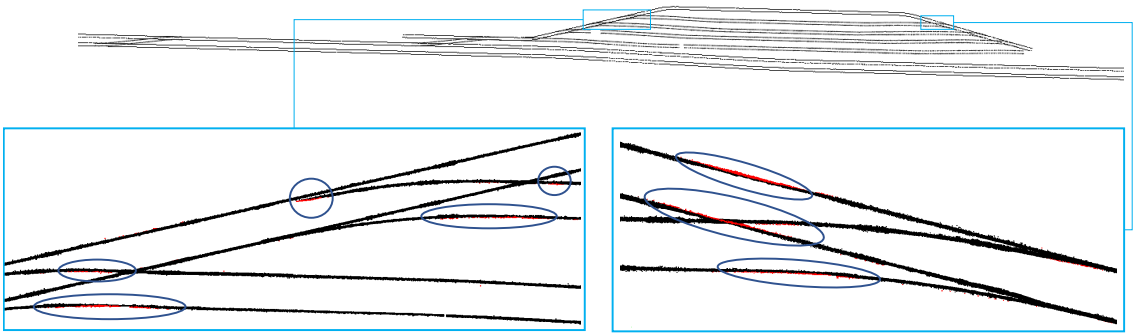


Fig. 9. True positives (black) and false negatives (red) within the extracted track head points.

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